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Global analysis of benthic complexity in shallow coral reefs

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Global analysis of benthic complexity in shallow coral reefs

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E-mail: gregasner@asu.edu**Keywords:** coral reef, benthic complexity, reef structure, fish biomass, marine protected areaSupplementary material for this article is available [online](#)

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Abstract

Three-dimensional shallow benthic complexity (also known as benthic rugosity) reflects the physical conditions of shallow coral reefs environments and can be used to estimate fish biomass and coral cover on reefs. Spatially explicit data on benthic complexity could offer critical information for coral reef conservation and management. However, benthic complexity has not yet been mapped at a global scale. We mapped global shallow water benthic complexity to 20 m depth at a spatial resolution of 10 m using 22 000 Sentinel-2 satellite images and a globally applicable underwater algorithm. We quantified geographic variation of benthic complexity in shallow coral reef areas from individual reef to ocean basin scales. We found that shallow benthic complexity is unevenly distributed worldwide, with high benthic complexity regions found in areas known to have high levels of benthic biodiversity such as the Coral Triangle, Coral Sea, and Great Barrier Reef. Yet nearly 60% of detected coral reef regions (size = 61 156 km²) are not listed as protected under current marine protected plans. These unprotected regions include substantial reef areas of high benthic complexity that may harbor high levels of biodiversity. Our global coral reef benthic complexity map supports plans to improve marine protected areas, reef conservation, and management.

1. Introduction

Shallow water (depth < 20 m) bottom spatial variation information (benthic complexity) reflects physical conditions of benthic environments, which plays a critical role in determining the distribution of marine biota [1–3]. It represents the three-dimensional (3D) physical structure of the seafloor, which mediates niche partitioning, habitat use, and trophic structure in corals, fishes, and non-coralline invertebrates [4, 5]. Spatially complex benthic habitats provide refuge for fishes from predators as well as suitable surfaces for coral settlement and growth to create coastal ecosystems with high biodiversity and fish biomass [6]. In addition, complex benthic habitats can help generate microclimates through physical interactions with temperature fields, water quality, underwater light density, wave action, and niches that can support high biodiversity [7].

On coral reefs and across associated aquatic environments, positive relationships between coastal

benthic complexity and coral cover as well as fish abundance, richness, and biomass have been well documented [8–11]. For example, previous studies demonstrate that benthic complexity measured at fine resolution relates to fish assemblage characteristics [9]. In addition, benthic complexity is often negatively related to algal cover, possibly as a result of high abundances of herbivores that moderate algal growth [12]. Benthic complexity may thus provide a proxy for biodiversity or fish assemblages which is a good indicator to be applied at a global scale.

Currently, no global coral reef benthic complexity assessments exist, and global spatial variation in benthic complexity remains unknown. Moreover, although many marine protected areas (MPA) have been designed worldwide, information is severely lacking on reef habitats (coral cover, rugosity, and biodiversity) covered by current protections [13]. Whether global conservation areas have covered the most benthic complexity by region is unknown. Most

benthic complexity estimations are derived at the scale of a single reef or within a small region because methods for detecting benthic complexity are spatially limited and based on localized data sources. Spatially explicit data on benthic complexity—a measure of habitat complexity—is needed at a global scale for coral reef conservation and management.

Over the past several decades, technologies have been developed that can support the mapping and monitoring of benthic complexity. These methods include field-based structure-from-motion photogrammetry, sounding, light detection and ranging (LiDAR), imaging spectroscopy, and multispectral satellite imagery [3, 9, 14–17]. Among these, field-based technologies (i.e. photogrammetry, ship-based sounding, airborne LiDAR) have limited spatial coverage and are time-consuming. In contrast, optical satellite imaging offers the unique advantage of estimating benthic complexity at both moderate spatial resolution (e.g. 10 m) and full spatial coverage. Furthermore, cloud computing platforms (computing network serves to process global-scale satellite data), like the Google Earth Engine (GEE), enable efficient processing of optical satellite images, opening a new pathway for mapping shallow reef benthic complexity at a global scale [18–21].

Our objective was to derive spatially-explicit information of global shallow (depth < 20 m) coral reef benthic complexity using optical satellite analysis methods. Mapped global benthic complexity reveals the 3D habitat of shallow coral reefs worldwide. We conducted a spatial analysis of global coral reef benthic complexity at different scales including ocean basins, country, Marine Ecoregions of the World (MEOW) boundaries, and MPA. We found nearly 60% of detected coral reef regions (size = 61 156 km²) are not listed as protected under current marine protected plans. These unprotected regions include substantial areas of high-benthic complexity coral reefs.

2. Materials and Methods

2.1. Global bathymetry mapping

We applied a global bathymetry mapping approach using the Sentinel-2 reflectance dataset in GEE [18]. Nearly 22 000 Sentinel-2 Level-2 A reflectance images (single image size: 100 km × 100 km) were processed in GEE to map bathymetry in global shallow reef regions. The derived bathymetry values ranged from 0.1 m to 20 m. In the processing, we first built a global clean water mosaic using the Reducer function in GEE. The Reducer function aggregated satellite data over selected time and space using statistics to calculate matrices. We selected Sentinel-2 images in coral reef regions with minimal cloud cover, cloud shadow, sunglint (sunlight spectral reflectance on the water surface), and water turbidity over two years

(i.e. 1 June 2018, to 1 June 2020) with multiple filters. We then aggregated all the resulting clean satellite images into a single global scale clean-water mosaic by taking median values of the pixel stack. We then applied a globally-applicable bathymetry estimation algorithm to derive bathymetry values for all shallow coral reef environments [18]. This algorithm was designed for clear coral reefs waters by comparing light attenuation levels between green and blue bands to estimate actual bathymetry (details could be found in Li *et al* [18]). After the bathymetry maps were derived, we applied the Global Coral Reef Extent map to mask out non-reef regions [22]. This reef map exhibits satellite-detected coral reef extents at a global scale. Therefore, only bathymetry maps in shallow coral reef areas were used to calculate benthic complexity.

2.2. Global benthic complexity mapping

Global shallow coral reef benthic complexity mapping is based on bathymetry data within coral reef extents. First, we filled missing pixels (less than 2% where no qualified satellite pixels could be found in two years) with the mean value of the nearest neighbor pixels. We then calculated slope and aspect values separately from bathymetry maps using GEE functions (*ee.Terrain.slope* and *ee.Terrain.aspect*). We examined multiple indices including ‘planar rugosity’, ‘vector ruggedness measure’, ‘land surface ruggedness index’, we found ‘planar rugosity’ was the best indicator to quantify benthic physical complexity which has the lowest noises and the highest relationship with abiotic and human drivers (details can be found in Asner *et al*) [14, 23, 24]. We calculated slope and aspect using GEE functions. After that, the benthic complexity index was calculated as: $\text{Benthic Complexity} = [\sin(\text{slope}) \times \sin(\text{aspect})]^2 + [\sin(\text{slope}) \times \cos(\text{aspect})]^2 + [\sin(\text{slope})]^2$. We post-processed the benthic complexity using a 2 × 2 pixel (20 × 20 m) moving circle-shaped boolean kernel (*ee.Kernel.circle*) to reduce noises. Therefore, the processed benthic complexity maps could illustrate different spatial variations of the benthic physical environment from reef to regional scales.

2.3. Geospatial statistics analyses

We calculated spatial zonal statistics using GEE functions (*image.reduceRegion*) at different scales (i.e. ocean basin, country, and MEOW). The spatial zonal statistics calculated the mean and standard deviation (SD) using all the benthic complexity pixels in the polygon boundary. For instance, a total of 251 450 000 benthic complexity pixels in the Great Barrier Reef and Warm Pool were used in the zonal statistics. We manually created region of interest (ROI) polygons for ocean basins. We also created the ROI of countries according to International Organization for Standardization (ISO) 3166:

The International Standard for country codes and codes for their subdivisions. This country classification is also used in the UNEP-WCMC database [25]. We used the MEOW polygons to calculate benthic complexity variations globally [26]. To analyze at a reef scale, we used benthic geomorphology boundaries from the Allen Coral Atlas project to calculate zonal statistics of benthic complexity [27]. The benthic geomorphology boundaries were classified as reef slope, reef crest, reef flat, plateau, patch reefs and lagoon. We calculated mean benthic complexity, SD of benthic complexity, histogram of benthic complexity, and reef area size at different scales. Then we calculated benthic complexity extent using (benthic complexity = mean benthic complexity value $\times$ reef area size). We used the following equation to calculate a hotspot index of benthic complexity (BC) extent (i.e. relative benthic complexity value) in each MEOW polygon

$$\frac{\text{BC extent of single MEOW polygon} - \text{mean BC extent of all MEOW polygons}}{\text{mean BC extent of all MEOW polygons}}$$

The World Database on Protected Areas (WDPA) database published in June 2021 was accessed in this study [13]. These marine WDPA polygons defined the current MPA. We compared compare benthic complexity information in both the MPA and non-MPA areas. We calculated mean benthic complexity, reef area, percentage of MPA area (Percentage = $\frac{\text{MPA area size}}{\text{Total area size}}$), and difference of benthic complexity in MPAs and non-MPAs areas (Differences of BC = MPA mean reef BC – Non MPA mean reef BC) in each MEOW polygon. We used a linear regression analysis to compare correlations between different indices (e.g. a correlation between benthic complexity and reef area). Also, benthic habitats map of the Allen Coral Atlas was introduced to analyze benthic complexity within different locations [27]. The Allen Coral Atlas benthic habitats map was classified as coral/algae, seagrass, microalgae mats, rock, rubble, and sand.

2.4. Validation

We used multiple methods to validate our derived benthic complexity results. Since benthic complexity is directly derived from the satellite derived bathymetry, we used field measured bathymetry sampling points to validate our satellite derived results. A total of 113 520 field sampling verification points were included, covering Heron Island, Australia; The Grenadines; Saona Island, Dominican Republic; Punta Cana, Dominican Republic; St. Croix, U.S. Virgin Islands; and Hawai'i Island [18]. We calculated the root mean square error (RMSE), Bias (m), and Root mean square (R^2) by using field measurements. We also accessed a large-scale benthic complexity data set measured by an airborne platform, Global Airborne Observatory (GAO) [14], to illustrate spatially comparisons of benthic complexity outputs.

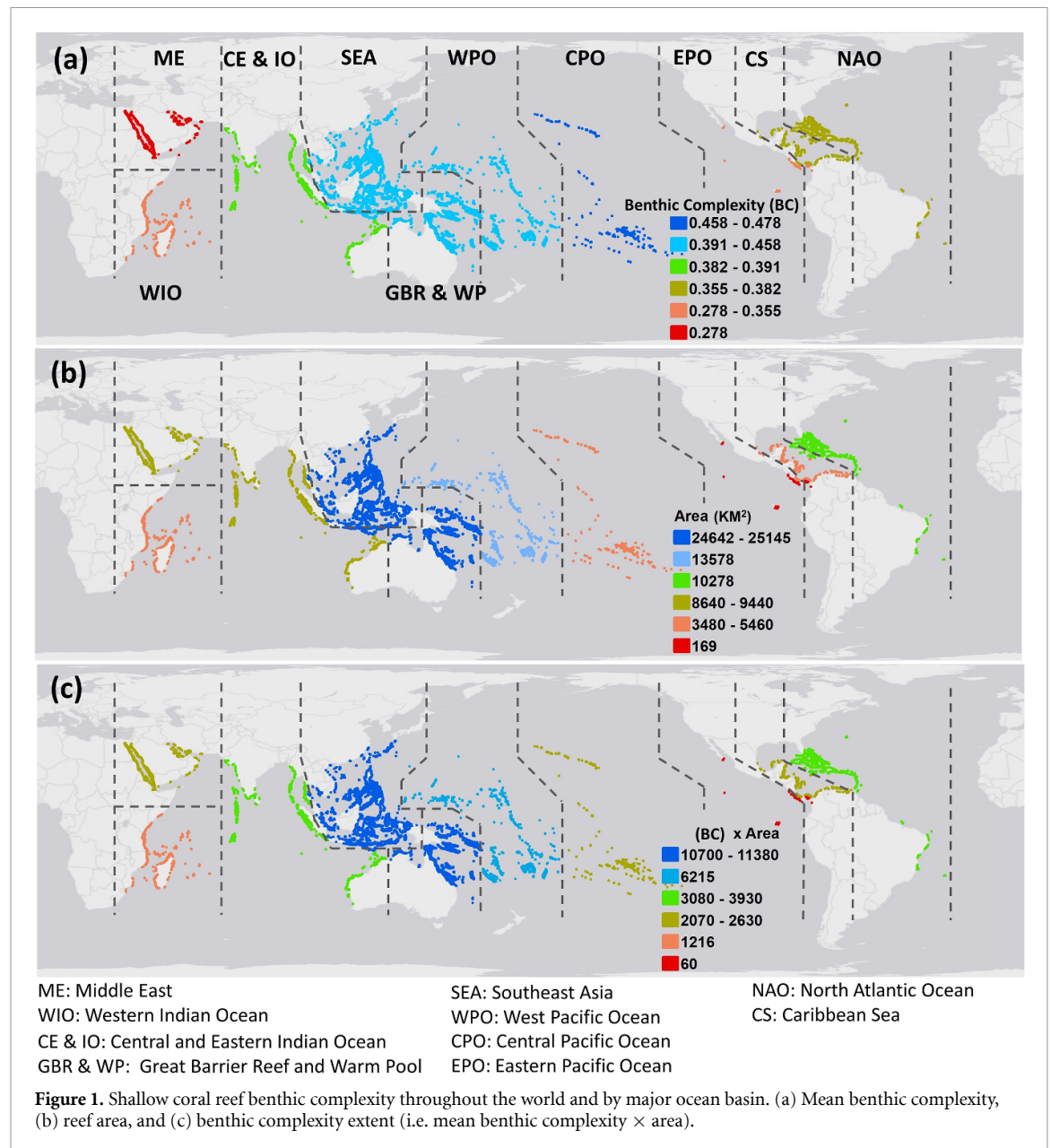
3. Results

3.1. Global coral reef benthic complexity variation

We detected spatial variation in mean benthic complexity among different ocean basins (figure 1, table S1). We found that the Middle East (ME) basin had the lowest mean benthic complexity worldwide of 0.278. All other basins had mean benthic complexity exceeding 0.34. Central Pacific Ocean (CPO, 0.478), West Pacific Ocean (WPO, 0.458), and South East Asia (SEA, 0.435) basins had the highest mean benthic complexity (figure S2). These represent mean benthic complexity variations in different ocean basins. Sentinel-2 detected contiguous shallow reef areas ranging in size from 169 km² to 25 145 km², depending on the region. In comparisons of benthic complexity histograms, basins (e.g. Great Barrier Reef and Warm Pool, West Pacific Ocean) generally have relatively even distributions. In contrast, the ME basin is dominated by low benthic complexity values of <0.3 (figures S1 and S2). The Great Barrier Reef and Warm Pool (GBR & WP), and South East Asia (SEA) basins, were comprised of the greatest reef area, with reefs exceeding 24 000 km² in size. West Pacific Ocean (WPO), North Atlantic Ocean (NAO), ME, and Central and Eastern Indian Ocean (CE & IO) basins had over 8500 km² of reef area. Despite its high benthic complexity, the Central Pacific Ocean basin had a relatively small reef area (4346 km²) (table S1). In contrast, the ME basin had a relatively large reef area (9430 km²), with a low mean benthic complexity value (0.278). In our comparison of benthic complexity extent (BC extent = mean benthic complexity value $\times$ reef area), the Great Barrier Reef and Warm Pool (GBR & WP, 11 384) as well as Southeast Asia (SEA, 10 722) basins had the largest expanses of the higher benthic complexity values than other basins (table S1). Since other basins had a mean value of 2696 and a SD of 1862.

3.2. Global coral reef benthic complexity analysis in MEOW

Mean benthic complexity values in different MEOW showed an uneven pattern, with the highest benthic complexity in the Pacific Ocean, Caribbean Ocean and Coral Triangle (figure 2, table S3). The MEOW regions had a mean benthic complexity value of 0.38 and an SD value of 0.11. Most (over 60%) MEOW regions had benthic complexity values ranging from 0.3 to 0.5 (figure S3). Reef regions in remote areas contained the highest benthic complexity values (table S3). For instance, Cocos-Keeling/Christmas Island, Vanuatu, Phoenix/Tokelau/Northern Cook Islands, and Northeast Sulawesi had mean benthic complexity values higher than 0.545 (the 95th percentile of all MEOW regions). Conversely, the lowest benthic complexity values were found in reef regions on or near a continental landmass. For instance,



Guianan, Western India, Western Arabian Sea, South India, Sri Lanka, Shark Bay, and Arabian (Persian) Gulf had mean benthic complexity values lower than 0.2 (the 5th percentile of all MEOW regions). Furthermore, using linear regression analysis to examine the correlation between reef area and mean benthic complexity in all MEOW regions, a low R square value was found ($R^2 = 0.05$) (figure S3). For instance, high benthic complexity was observed in both small islands (e.g. Cocos-Keeling/Christmas Island, Phoenix/Tokelau/Northern Cook Islands) and large-area reef regions (e.g. Central and Southern Great Barrier Reef, Bahamian, Solomon Sea). Benthic complexity extent ranged from 0.056 to 3063. MEOW regions have a mean value of benthic complexity extent of 468 and an SD value of 688. In general, large MEOW regions (area $> 2000 \text{ km}^2$) had relatively high [>0.38 (mean value)] benthic

complexity, which led to high benthic complexity extents (benthic complexity extent > 1000). Limited by small reef areas, some high-benthic complexity islands showed low benthic complexity extents (e.g. Cocos-Keeling/Christmas Island < 2.29 , Rapa-Pitcairn < 6.89).

We identified regional hotspots where relative benthic complexity extent was the highest (figure 2(c), table S3). The highest hotspots regions represented both high benthic complexity and large reef area. The hotspot index ranged from -1 to 6.75 with a mean value of 0.0 and an SD value of 1.47 . Central and Southern Great Barrier Reef, Torres Strait Northern Great Barrier Reef and Palawan/North Borneo had the highest benthic complexity hotspot index at >2.84 (the 95th percentile of all MEOW regions). Especially, Central and Southern Great Barrier Reef have the highest hotspots index of 6.75 . In

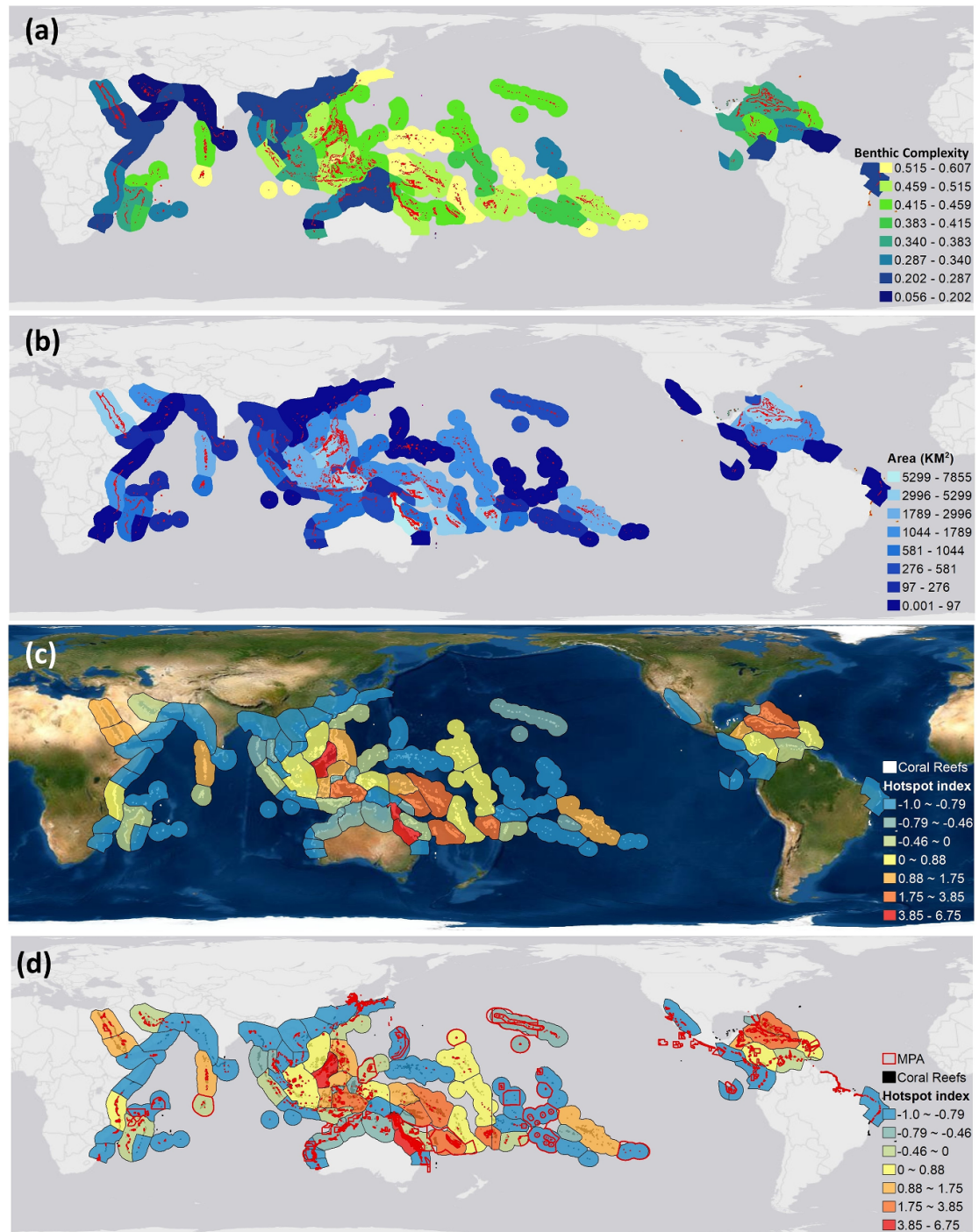


Figure 2. Shallow coral reef benthic complexity in Marine Ecoregions of the World (MEOW) regions. (a) Mean benthic complexity, (b) reef area, (c) benthic complexity hotspot index, and (d) comparison between benthic complexity hotspot index and marine protected area (MPA).

the analysis of global spatial hot spots index patterns, we found that hotspot regions were spatially concentrated. The major high hot spot regions concentrated in the Coral Triangle, Coral Sea, and Great Barrier Reef regions and the Caribbean regions (figure 2(c)).

3.3. Benthic complexity extent ranked by country

Countries displayed distinct variations in benthic complexity extents (table S2). Over 70% of global benthic complexity extents were concentrated in 10

countries that each had benthic complexity extents higher than 1200. Among these, Australia had the highest benthic complexity extent of 7483, which was 17.6% of global benthic complexity extent. Indonesia (5865), Islands of France (3240), Papua New Guinea (3182), and the Philippines (3064) also had high benthic complexity extents. These regions were followed by Fiji (1887), The Bahamas (1546), Saudi Arabia (1508), Solomon Islands (1437), and the Maldives (1284). All other countries and their subdivisions

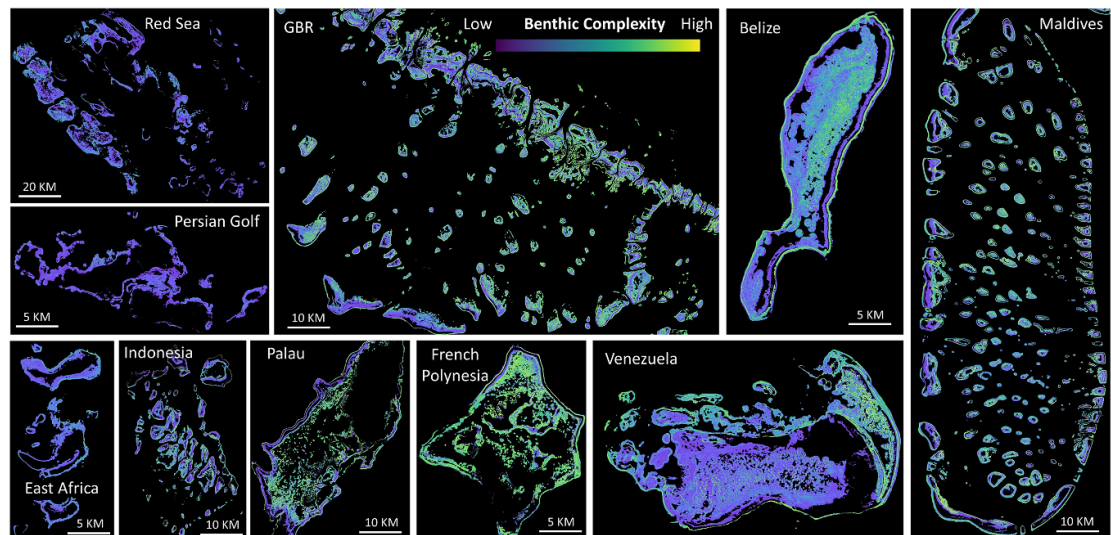


Figure 3. High (10 m) spatial resolution benthic complexity variation at the intra- and inter-reef scales selected from our global mapping results.

shared 30% of global coral reef benthic complexity extents, with most countries containing values <250.

3.4. MPA assessment using benthic complexity

Nearly 59% of global reef area (size = 61 156 km²) detected by Sentinel-2 was not incorporated into a MPA. MPAs were unevenly distributed across 91 global MEOW regions (figure 2(d), table S3). MPAs coverage ranged from 0% to 100% with a mean value of 37% and an SD value of 33%. Total of 28 MEOW regions had <10% MPA coverage. Some low (<10%) MPA coverage regions had high benthic complexity extents (e.g. Solomon Sea: 1804, Bahamian: 1675, Solomon Archipelago: 1505, Tuamotus: 1264, Maldives: 1288, Eastern Philippines: 1274). In contrast, 20 MEOW regions had > 67% (the 80th percentile) MPA coverage. These regions contributed over 50% of the global MPA area and included regions with very high benthic complexity extents (e.g. Central and Southern Great Barrier Reef: 7855, Torres Strait Northern Great Barrier Reef: 4565, New Caledonia: 2456). We calculated the difference between MPA benthic complexity and Non-MPA benthic complexity (MPA BC—Non MPA BC) (table S3). We found in a total of 52 MEOW regions, areas not covered by MPAs had higher benthic complexity values than areas not covered by MPAs. Even in some high-MPA-coverage [MPA coverage > 67% (the 80th percentile)] MEOW regions (e.g. South Kuroshio, Gulf of Papua, Southwestern Caribbean), high reef benthic complexity areas were not included in MPAs.

3.5. Benthic complexity patterns at a reef scale

Our benthic complexity maps revealed high spatial variation within individual reefs (figure 3). Benthic complexity values varied widely from the inner reef area to reef slope (figure S4). On reef slopes, benthic

complexity values were usually higher than in relatively flat inner-reef areas. For instance, in the typical reef regions including Central and Southern Great Barrier Reef, Fiji Islands, Maldives, Western Caribbean, and Coral Sea, the highest benthic complexity values on a reef are 0.72–0.81 in reef slope regions with a mean value of 0.78. In contrast, the lowest benthic complexity values are 0.12–0.16, which are often observed in flat inner-reef areas with a mean value of 0.13. Highly complex coral habitats had higher benthic complexity values than relatively sandy (dominated by sand with other benthic habitats) and hard-bottom areas (figure S4). Different reefs also showed different reef benthic complexity values. For instance, reefs in the Red Sea and Persian Gulf had much lower benthic complexity compared to reefs in other regions (figure 3).

3.6. Benthic complexity validation

We compared bathymetry point values from satellite-derived bathymetry maps with field measurements. We calculated RMSE values varying from 1.26 to 1.98 m. Bias results ranged from 0.83 to 1.09 m, and R^2 values were 0.78–0.98. The satellite-derived bathymetry offered reasonable results for retrieving benthic complexity. Detailed bathymetry validation results can be found in Li *et al* (2021). We plotted benthic complexity maps derived from Sentinel-2 (10 m resolution) and GAO airborne (2 m resolution) (figure S5) [14]. In comparing spatial variation of benthic complexity, Sentinel-2 derived results provide the same spatial variations as in GAO measurements. Using benthic complexity maps and benthic habitats maps, we found on the low slope sandy dominated reef area of the island of Moloka'i, both Sentinel-2 and GAO measured low benthic complexity values (figure S5). On the high complexity coral habitats areas of

the island of O'ahu, both maps accurately quantified benthic complexity (figure S5). However, Sentinel-2 lost some finer details in comparison to GAO results due to the relatively coarse spatial resolution (10 m).

4. Discussion

Using benthic complexity results in different MEOW regions, we found distinct differences in benthic complexity between remote islands, which typically had high benthic complexity, and reefs adjacent to the mainland, which had low benthic complexity (figure 2, tables S2 and S3). This pattern indicates that the terrestrial environment likely affects shallow coastal zone ecosystems including the benthic habitat complexity [28, 29]. The terrestrial environment generates multiple inputs to coastal reefs including sediments, nutrients, and other satellite detectable components (e.g. colored dissolved organic matter) through river plumes and coastal surface runoff [30–32]. Highly turbid rivers deposit sediments onto reefs that can reduce benthic surface roughness, a phenomenon that likely affects Western India, where we observed very low benthic complexity [33]. Coastal development and human activities may also put stress on the coral reef environment [34, 35]. Continental landmasses can impact reefs through terrestrial inputs from land to sea (e.g. runoff from nearshore development), and by increasing human access (fishing, trampling, etc) to reefs [36–38]. Our global reef benthic complexity map may help to identify highly vulnerable regions with greater land-based impacts that might benefit from conservation and management. Actionable management plans need to be developed to mitigate land-to-sea effects on reefs to sustain benthic complexity and habitat function.

Our global coral reef complexity maps reveal nested patterns of benthic habitat variation, from individual reef to ocean basin levels, with implications for conservation, management, and marine policy. Spatial distributions of benthic complexity provide information for MPA planning and development in the future. Today, MPA boundaries are planned on inconsistent criteria, and about 60% of global reef area remains unprotected. Unprotected areas include many large reef regions with high benthic complexity (e.g. Solomon Sea, Bahamian, Solomon Archipelago, Tuamotus, Maldives, Eastern Philippines). Our maps can support more effective MPA planning by indicating areas with high benthic complexity, which in many regions is a proxy for benthic biodiversity [39, 40]. A potential pathway to creating efficient coral reef conservation regions is to combine global benthic complexity, global bathymetry, and global benthic habitat maps to locate areas of high benthic habitat diversity and habitat complexity [18, 22, 27]. For instance, the Red Sea region has enormous reef areas with mixed low and

high benthic complexity values. Marine conservation area plans could be developed at a reef with high biodiversity to increase management efficiency [41]. Also, efficient coral management could be conducted in high benthic complexity regions to avoid places that have an extremely low benthic complexity. Some regions with modest protections (e.g. South Kurashio, Gulf of Papua, Southwestern Caribbean) could consider adding areas with high benthic complexity. Benthic complexity can generally be used as an index to locate a region (e.g. low terrestrial impacts, diverse habitats) for effective conservation and management efforts.

A previous study showed that benthic complexity is dominantly driven by bathymetry and slope [14]. Specifically, water depth and benthic slope shaped underwater physical conditions and affect subsidence, erosion, and other factors affecting underwater benthic physical structures [42, 43]. Therefore, field-measured bathymetry data is a good indicator to validate our benthic complexity results. In the meantime, in comparing Sentinel-2 derived benthic complexity with a successfully deployed GAO measured benthic complexity [14], the Sentinel-2 map successfully revealed detailed spatial variations of benthic complexity at an individual reef scale in Hawaiian Islands (figure S5). However, Sentinel-2 imagery cannot detect small coral colonies and rock outcroppings that dictate fish and invertebrate dynamics at an ultra-fine scale (e.g. 40–60 cm), which exceeds satellite detection ability [14, 44, 45]. The 10 m resolution benthic complexity maps represent the highest resolution available at a global scale. Other finer-resolution satellite images have drawbacks in global mapping; for instance, we found that Planet Dove global mosaics at 4 m spatial resolution could not derive suitable bathymetry maps for estimating benthic complexity, since Planet's mosaic processing modifies the spectral properties of the benthos in blue and green spectral bands [46, 47]. In addition, multi-spectral satellite-derived reef benthic complexity has a bathymetry detection limitation (i.e. 20 m depth in this study). Because deep (>20 m) waters attenuate the majority of sunlight, deep waters lead to greater uncertainty in bathymetry detection [18].

Here we mapped global shallow water coral reef benthic complexity using 22 000 Sentinel-2 satellite images at a 10 m spatial resolution using a newly developed, globally applicable, and validated method. We found that shallow benthic complexity is unevenly distributed worldwide. Moreover, nearly 60% of detected coral reef regions (size = 61 156 km²) are not listed as protected under current marine protected plans. These unprotected regions include substantial areas of high-benthic complexity coral reefs that may harbor high levels of biodiversity. Our global benthic complexity map could be combined with other global maps (e.g. Allen Coral Atlas benthic/geomorphology map, UNEP seagrass map, etc) to advance the design

of biologically and ecologically effective MPA and other management areas, and to support innovations that improve reef resilience.

Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

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